

Problem 2

Computing the integral for the total radiation

In[44]:=

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Integrate[
  ((1 - β * Cos[θ])^2 - (1 - β^2) * (Sin[θ])^2 * (Cos[φ])^2) /
  ((1 - β * Cos[θ])^5), {θ, 0, Pi}, {φ, 0, 2 * Pi}]
```

Out[44]=

$$\frac{\pi^2 \sqrt{\frac{1}{1-\beta}} (12 + 7\beta^2)}{8(-1+\beta)^2(1+\beta)^{5/2}} \text{ if } \text{condition} +$$

In[45]:=

```
FullSimplify[%]
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$$\frac{\pi^2 \sqrt{\frac{1}{1-\beta}} (12 + 7\beta^2)}{8(-1+\beta)^2(1+\beta)^{5/2}} \text{ if } \text{condition} -$$

(β ∉ ℝ || -1 ≤ Re[β] ≤ 1) &&

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(β ∉ ℝ || -1 < Re[β] < 0 || 0 < Re[β] < 1) &&

(ArcCsc[β] ∉ ℝ || Re[ArcSec[β]] < 0 || Re[ArcSec[β]] > π)

Below we plot the required radiation. At first we plot what we got for this problem (0.5 and 0.2 beta values), then we will plot what we found for Bremstrahlung.

In[47]:=

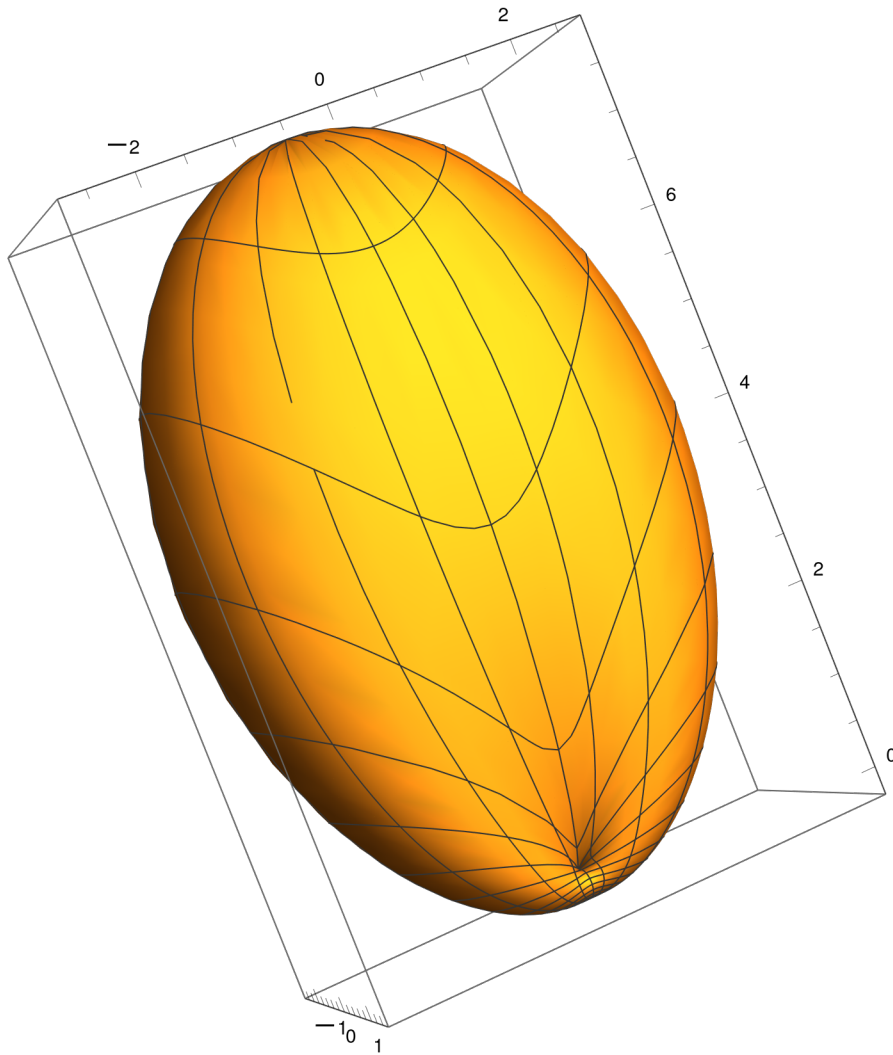
 $\beta = 0.5$

$$\text{SphericalPlot3D}\left[\frac{\left((1 - \beta \cos[\theta])^2 - (1 - \beta^2) \sin^2[\theta] \cos^2[\phi]\right)}{(1 - \beta \cos[\theta])^5}, \{\theta, 0, \text{Pi}\}, \{\phi, 0, 2 * \text{Pi}\}\right]$$

Out[47]=

0.5

Out[48]=



In[51]:=

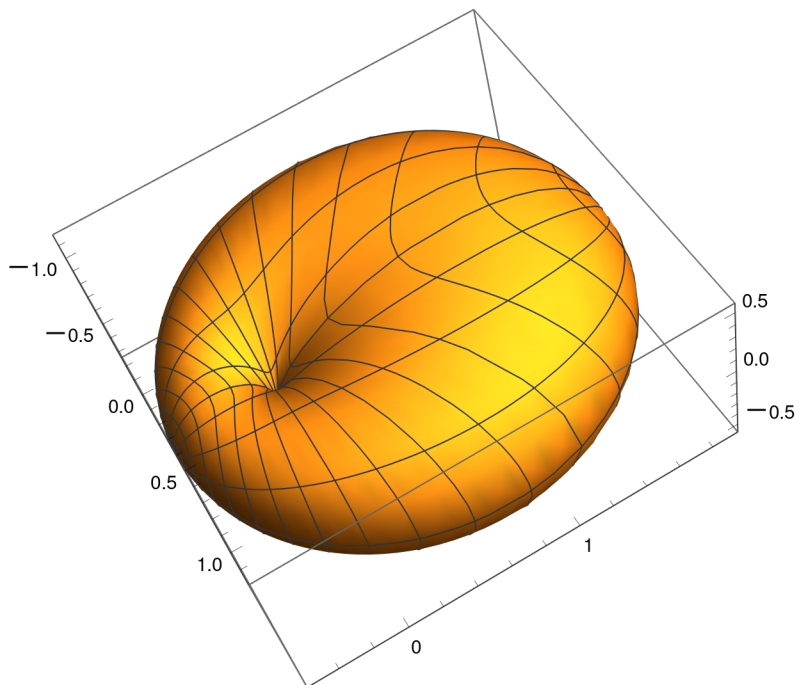
 $\beta = 0.2$

$$\text{SphericalPlot3D}\left[\frac{\left((1 - \beta \cos[\theta])^2 - (1 - \beta^2) \sin^2[\theta] \cos^2[\phi]\right)}{(1 - \beta \cos[\theta])^5}, \{\theta, 0, \text{Pi}\}, \{\phi, 0, 2 * \text{Pi}\}\right]$$

Out[51]=

0.2

Out[52]=



For the Bremstrahlung (beta 0.5 and 0.2)

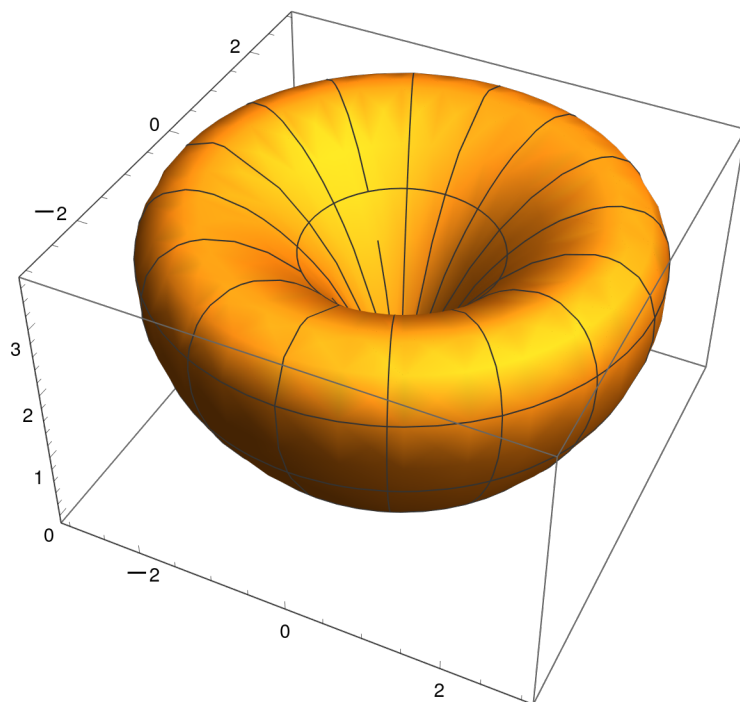
In[53]:=

 $\beta = 0.5$
$$\text{SphericalPlot3D}\left[\frac{\sin^2[\theta]}{(1 - \beta \cos[\theta])^5}, \{\theta, 0, \pi\}, \{\phi, 0, 2\pi\}\right]$$

Out[53]=

0.5

Out[54]=



In[55]:=

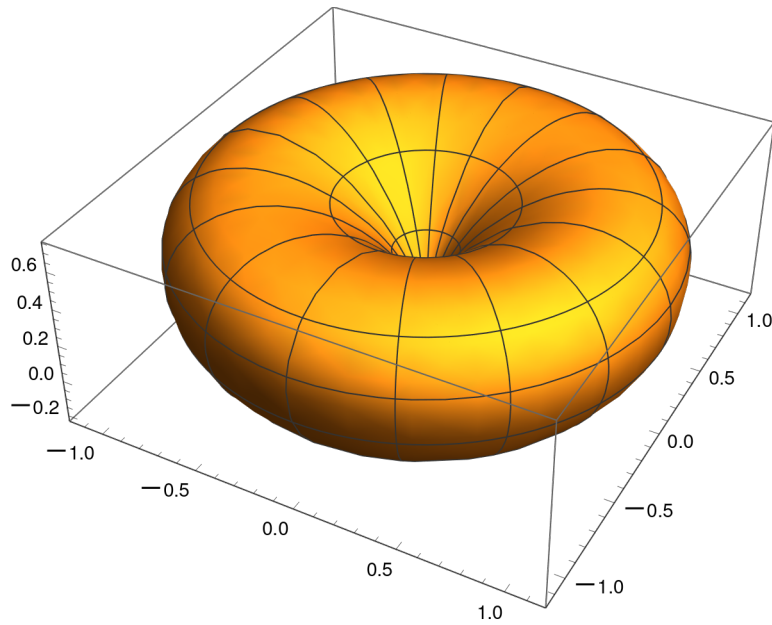
$$\beta = 0.2$$

SphericalPlot3D[(Sin[θ]²) / ((1 - β * Cos[θ])⁵), { θ , 0, Pi}, { ϕ , 0, 2 * Pi}]

Out[55]=

0.2

Out[56]=



Problem 6 (a)

In[30]:=

$$Ex' = Ex$$

$$Bx' = Bx$$

$$Ey' = \gamma * (Ey - v * Bz)$$

$$By' = \gamma * (By + (v / c^2) * Ez)$$

$$Ez' = \gamma * (Ez + v * By)$$

$$Bz' = \gamma * (Bz - (v / c^2) * Ey)$$

$$\gamma = 1 / \text{Sqrt}[1 - (v^2 / c^2)]$$

Out[30]=

$$Ex$$

Out[31]=

$$Bx$$

Out[32]=

$$\frac{Ey - Bz v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Out[33]=

$$\frac{By + \frac{Ez v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Out[34]=

$$\frac{Ez + By v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Out[35]=

$$\frac{Bz - \frac{Ey v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Out[36]=

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Below we compute the expanded form of $\text{vec}(E) \cdot \text{vec}(B)$

In[39]:=

$$Ex' \cdot Bx' + Ey' \cdot By' + Ez' \cdot Bz'$$

Out[39]=

$$Bx Ex + \frac{\left(Ez + By v\right) \left(Bz - \frac{Ey v}{c^2}\right)}{1 - \frac{v^2}{c^2}} + \frac{\left(Ey - Bz v\right) \left(By + \frac{Ez v}{c^2}\right)}{1 - \frac{v^2}{c^2}}$$

In[38]:=

$$\text{FullSimplify}\left[Ex' \cdot Bx' + Ey' \cdot By' + Ez' \cdot Bz'\right]$$

Out[38]=

$$Bx Ex + By Ey + Bz Ez$$

We verified $E \cdot B$ is invariant.

Problem 6 (b)

First let's compute $E'^2 - c^2 B'^2$

In[41]:=

$$\left(Ex' \cdot Ex' + Ey' \cdot Ey' + Ez' \cdot Ez'\right) - c^2 \left(Bx' \cdot Bx' + By' \cdot By' + Bz' \cdot Bz'\right)$$

Out[41]=

$$Ex^2 + \frac{\left(Ez + By v\right)^2}{1 - \frac{v^2}{c^2}} + \frac{\left(Ey - Bz v\right)^2}{1 - \frac{v^2}{c^2}} - c^2 \left(Bx^2 + \frac{\left(Bz - \frac{Ey v}{c^2}\right)^2}{1 - \frac{v^2}{c^2}} + \frac{\left(By + \frac{Ez v}{c^2}\right)^2}{1 - \frac{v^2}{c^2}}\right)$$

In[40]:=

$$\text{FullSimplify}\left[\left(Ex' \cdot Ex' + Ey' \cdot Ey' + Ez' \cdot Ez'\right) - c^2 \left(Bx' \cdot Bx' + By' \cdot By' + Bz' \cdot Bz'\right)\right]$$

Out[40]=

$$-\left((Bx^2 + By^2 + Bz^2) c^2\right) + Ex^2 + Ey^2 + Ez^2$$

Full Simplification shows that we have gotten the non-prime factors of E_i , B_j intact, that means that the spacetime-interval like quantity $E^2 - c^2 B^2$ is invariant.

Problem 6 (c)

Answer: No

Reason: For one system if $B=0$ then $S^2 = E^2 - c^2 B^2 = E^2 > 0$ which implies $S^2 > 0$
But now for another system where $E = 0$ we get $S^2 = -c^2 B^2$ which implies $S^2 < 0$

But we know that S^2 is conserved for all frames, hence it can't be greater and lower than zero for the same system in two different frames. This means the answer to the question is NO.